Shayan Sadigh

Location: Santa Clara, CA
Email: ssadigh.contact@gmail.com
Github: https://github.com/shayanpersonal (200+ stars)

Summary

- AI Engineer | Previously @Amazon SageMaker
- Placed top-10 in $1.5 million competition sponsored by Department of Homeland Security. Designed and trained ~100 million parameter MVCNN neural network for threat recognition. (Slides)
- Authored "checkered subsampling" deep learning research paper. (arXiv)
- Ranked first globally (tied) in competitive arena shooter.

Work Experience

aurascape.ai
AI Engineer (Feb. 2024 - Current)

- Addressing security challenges with LLM agents. Founding team.

Santa Barbara City College (SBCC)
Adjunct Professor of Computer Science (Aug. 2023 - Dec. 2023)

- Walked students through various aspects of designing and training AI / deep learning models as part of semester-long in-person course. Discussed linear layers, nonlinearities, normalization, convolution, aspects of transformers such as attention layers and embeddings. Discussed backpropagation and loss functions.
- Assigned k-nearest-neighbors and neural nets including a simple "small language model" in PyTorch for homework. Trained models on MNIST and tiny-shakespeare. Used MoeBot as an AI teaching assistant.

moebot.ai
AI Engineer (2021-2023)

- Owner of moebot.ai. Built subscription-based LLM agent for Discord servers. Takes discrete actions such as deciding whether to respond or send emoji reactions via prompt engineering. Stores and recalls rudimentary "memory" of users (a form of RAG or Retrieval Augmented Generation). Runs on Google Cloud, Redis, Flask, Python. Generates revenue through Stripe transactions.
- Used MoeBot as an AI teaching assistant for a college course. Built various smaller projects.

Amazon Web Services (AWS)
Deep Learning Engineer II (Mar. 2020 - Dec. 2020)

- Helped integrate Apache TVM deep learning compiler into Amazon SageMaker's PyTorch fork.

Palo Alto Networks
Senior Software Engineer (Oct. 2019 - Mar. 2020)
Software Engineer (Sep. 2018 - Sep. 2019)

- Awards: Project of the Quarter - Machine Learning Antivirus
- Proposed and implemented new stack-based pattern matching algorithm which improved threat coverage in PAN-OS for a certain class of threats, written in C.
- Helped integrate machine learning models into resource-constrained firewall.
- Brought up and helped train two junior engineers.

University of California, Santa Barbara
Teaching Assistant (Jan. 2018 - Mar. 2018)

- Teaching assistant for CS176A: Introduction to Computer Networks.

Palo Alto Networks
Software Engineer Internship (Jun. 2017 - Sep. 2017)

- Developed software tools for processing TLS 1.3 encrypted traffic and PCAPNG packet captures.
- Performed various lab maintenance tasks such as bringing up ESXI servers / Palo Alto Networks firewalls.

Projects

- stacked-autoencoder-pytorch (100+ stars) - a popular PyTorch autoencoder.
- Checkered CNN (30+ stars) - my deep learning technique, checkered subsampling.
- Passenger Screening Model (20+ stars) - solution to $1.5 million passenger screening challenge.
- GauchoMap Chrome Extension - 200+ users when I was maintaining it (2016).

Education

- 5-year accelerated B.S. + M.S. program @ University of California, Santa Barbara
- Master of Computer Science (2018)
- Bachelor of Computer Science (2017)

Miscellaneous

- Enjoys playing piano / making music.
- Expert Dance Dance Revolution player.
- 2nd-degree black belt in Soo Bahk Do.
- Languages: English (native), Farsi (native, but very rusty)